

How to derive a CI?

Main idea:

- Collect a random sample
- Determine the parameter of interest, say θ
- Find a random variable, say Y , that involves parameter θ and the random sample.
- Specify the distribution of Y .
- Use the distribution of Y to find random lower and upper bounds on θ

Parameter of interest: true mean, μ

- $X_1, \dots, X_n$: A random sample with $E[X] = \mu$
- We are looking for a random variable $Y = h(X_1, \dots, X_n; \mu)$ whose distribution can be specified.

	X is normal		X is not normal		
	$n < 40$	$n \geq 40$	$n < 30$	$30 \leq n < 40$	$n \geq 40$
σ : known	$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$	$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$	?	$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$	$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$
σ : unknown	$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$	$Z = \frac{\bar{X} - \mu}{S/\sqrt{n}}$	?	?	$Z = \frac{\bar{X} - \mu}{S/\sqrt{n}}$

CIs around the true mean, μ

- $X_1, \dots, X_n$: A random sample with $E[X] = \mu$
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	X is normal		X is not normal		
	$n < 40$	$n \geq 40$	$n < 30$	$30 \leq n < 40$	$n \geq 40$
σ : known	$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$	$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$	?	$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$	$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$
σ : unknown	$\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$	$\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$	?	?	$\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$

Parameter of interest: true proportion, p

X : Number of successes in n trials, each trial is i.i.d. with constant $p = P\{\text{success}\}$

We are looking for a random variable, $Y = h(X; \theta)$ whose distribution can be specified.

$$\hat{p} = \frac{X}{n}$$

$$\tilde{p} = \frac{X + 2}{n + 4}$$

$$\tilde{n} = n + 4$$

$n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$	No condition on X or n
$Z = \frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}}$	$Z = \frac{\tilde{p} - p}{\sqrt{\frac{\tilde{p}(1 - \tilde{p})}{n}}}$

CIs around the true population proportion p

X : Number of successes in n trials, each trial is i.i.d. with constant $p = P\{\text{success}\}$

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$$\hat{p} = \frac{X}{n}$$

$$\tilde{p} = \frac{X + 2}{n + 4}$$

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$$n\hat{p} \geq 10 \text{ and } n(1 - \hat{p}) \geq 10$$

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

No condition on n or p

$$\tilde{p} \pm z_{\alpha/2} \sqrt{\frac{\tilde{p}(1 - \tilde{p})}{\tilde{n}}}$$

A Confidence Interval for the Difference Between Two Means

$X_1, \dots, X_{n_X}$: a random sample from a population with μ_X & σ_X^2
 $Y_1, \dots, Y_{n_Y}$: a random sample from a population with μ_Y & σ_Y^2

X & Y samples
are independent
of one another.

Samples: normal
 σ_X and σ_Y : known

100(1 - α)% CI
for $\mu_X - \mu_Y$ is:

$$\bar{X} - \bar{Y} \pm z_{\alpha/2} \sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}$$

Samples: any distribution
 σ_X and σ_Y : unknown
 n_X and n_Y : large

100(1 - α)% CI
for $\mu_X - \mu_Y$ is:

$$\bar{X} - \bar{Y} \pm z_{\alpha/2} \sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}$$

$$Z = \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}}$$

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Normal samples, unknown variances, small sample sizes

$X_1, \dots, X_{n_X}$: a random sample from a population with μ_X & σ_X^2

$Y_1, \dots, Y_{n_Y}$: a random sample from a population with μ_Y & σ_Y^2

X & Y samples are independent of one another.

$$\frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}} \sim t_v$$

$$v = \left\lfloor \frac{\left(\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}\right)^2}{\frac{\left(\frac{s_X^2}{n_X}\right)^2}{n_X - 1} + \frac{\left(\frac{s_Y^2}{n_Y}\right)^2}{n_Y - 1}} \right\rfloor$$

Confidence Intervals for the Difference Between Two Proportions

Traditional approach?

New approach?

$$Z = \frac{\tilde{P}_X - \tilde{P}_Y - (p_X - p_Y)}{\sqrt{\frac{\tilde{p}_X(1 - \tilde{p}_X)}{\tilde{n}_X} + \frac{\tilde{p}_Y(1 - \tilde{p}_Y)}{\tilde{n}_Y}}}$$

$$\tilde{P}_X = \frac{X+1}{\tilde{n}_X}, \tilde{n}_X = n_X + 2, \tilde{P}_Y = \frac{Y+1}{\tilde{n}_Y}, \tilde{n}_Y = n_Y + 2$$

100(1- α) Confidence Interval around $p_X - p_Y$ New Approach

100(1 - α)% CI for the difference $p_X - p_Y$ is

$$\tilde{p}_X - \tilde{p}_Y \pm z_{\alpha/2} \sqrt{\frac{\tilde{p}_X(1 - \tilde{p}_X)}{\tilde{n}_X} + \frac{\tilde{p}_Y(1 - \tilde{p}_Y)}{\tilde{n}_Y}}$$

If the lower limit of the confidence interval is less than -1, replace it with -1.

If the upper limit of the confidence interval is greater than 1, replace it with 1.

$$\tilde{P}_X = \frac{X+1}{\tilde{n}_X}, \tilde{n}_X = n_X + 2, \tilde{P}_Y = \frac{Y+1}{\tilde{n}_Y}, \tilde{n}_Y = n_Y + 2$$

$$\tilde{p}_X - \tilde{p}_Y \pm z_{\alpha/2} \sqrt{\frac{\tilde{p}_X(1 - \tilde{p}_X)}{\tilde{n}_X} + \frac{\tilde{p}_Y(1 - \tilde{p}_Y)}{\tilde{n}_Y}}$$

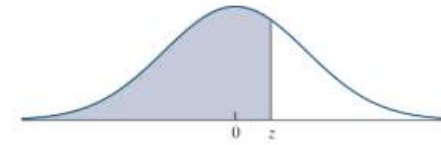
[Popcorn-poll.xlsx](#)

Example

The production manager wants to compare the average weights of popcorns from two production lines.

The quality department checks the number of popcorns within the specified weight limits. The manager would like to know whether the true proportions of popcorns in the limits are different than each other. Let X_i denote the number of popcorns within specifications. He obtained the following data:

Sample	n	X_i
Line 1	12	11
Line 2	16	14



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952

CIs around p

99%	95%	90%
(0, 0.359)	(0, 0.279)	(0, 0.238)

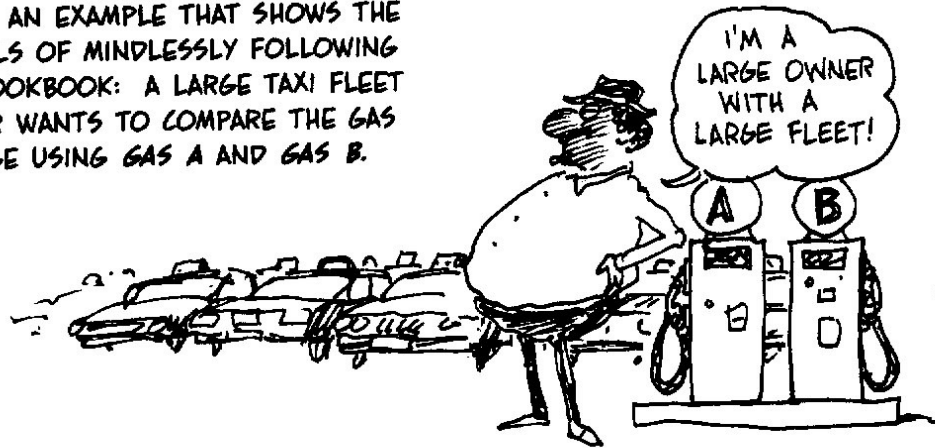
Are the true proportions of popcorns in the limits different than each other?

Paired Data: Main Idea

Same shorts,
Same person, 122
kg to 61 kg, in just
over one year



HERE'S AN EXAMPLE THAT SHOWS THE PITFALLS OF MINDLESSLY FOLLOWING THE COOKBOOK: A LARGE TAXI FLEET OWNER WANTS TO COMPARE THE GAS MILEAGE USING GAS A AND GAS B.



How to use or not to use statistics?

STARTING WITH 100 CABS, HE RANDOMLY ASSIGNS 50 TO EACH GASOLINE, AND, AFTER A DAY'S DRIVING, DETERMINES

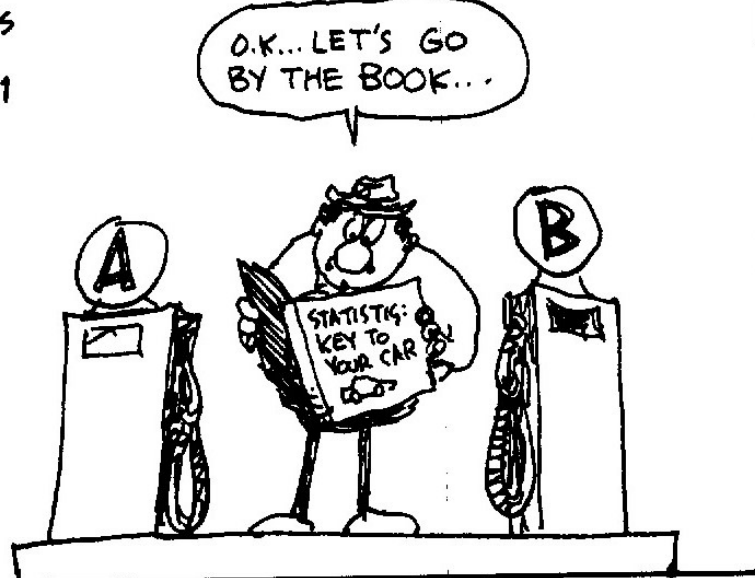
	SAMPLE SIZE	MEAN MILEAGE	STANDARD DEVIATION
A	50	25	5.00
B	50	26	4.00



THE SAMPLE DIFFERENCE IS

$$\bar{x}_1 - \bar{x}_2 = 25 - 26 = -1$$

IS GAS B REALLY BETTER THAN GAS A?



OWING TO THE LARGE STANDARD DEVIATIONS, THE STANDARD ERROR IS PRETTY SUBSTANTIAL:

$$\begin{aligned} SE(\bar{X}_1 - \bar{X}_2) &= \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \\ &= \sqrt{\frac{25}{50} + \frac{16}{50}} \\ &= .905 \end{aligned}$$

AT THE 95% CONFIDENCE LEVEL, WE HAVE

$$\begin{aligned} \mu_1 - \mu_2 &= \bar{x}_1 - \bar{x}_2 \pm z_{.025}(.905) \\ &= -1 \pm (1.96)(.905) \\ &= -1 \pm 1.774 \end{aligned}$$

THIS INCLUDES THE VALUE 0, CORRESPONDING TO $\mu_1 = \mu_2$



THIS EXCEEDS THE $\alpha = .05$ SIGNIFICANCE LEVEL, SO WE CONCLUDE THAT THE EVIDENCE IN FAVOR OF EITHER GAS IS VERY WEAK.



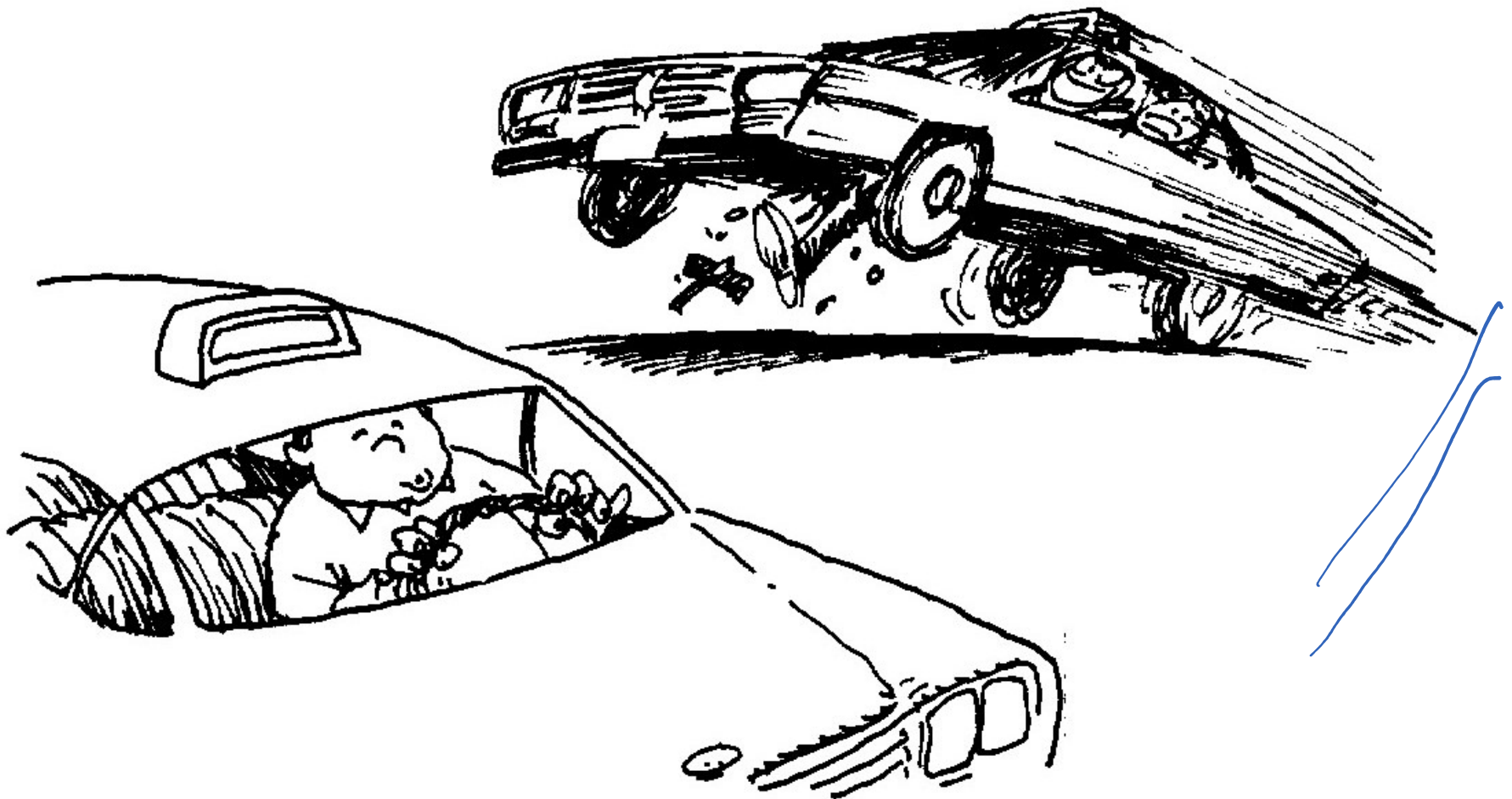
PAIRED COMPARISONS

a better way to compare gasolines

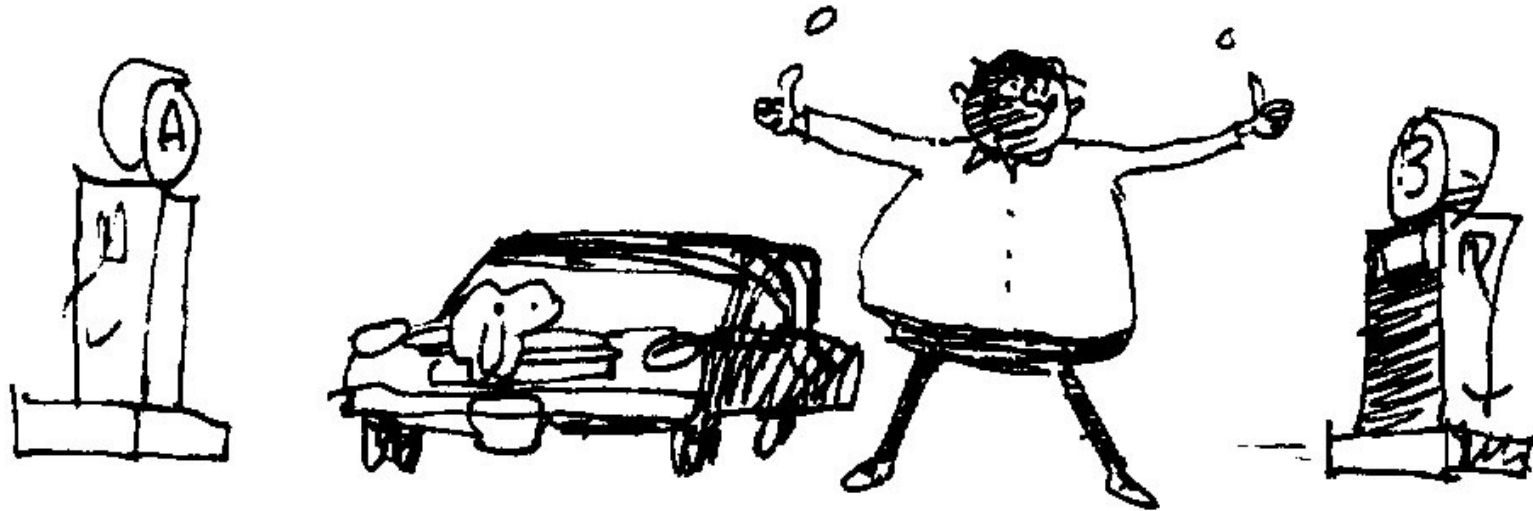


THE TAXI OWNER FOLLOWED THE COOKBOOK EXACTLY. HIS SAMPLES WERE RANDOM, AND HIS SAMPLE SIZE WAS LARGE ENOUGH. HE JUST FAILED TO THINK WHEN NECESSARY!

ALTHOUGH GAS B APPEARS TO BE SLIGHTLY BETTER THAN GAS A, THE CONFIDENCE INTERVAL WAS WIDE BECAUSE OF THE LARGE STANDARD DEVIATIONS—I.E., THE MILEAGES VARIED WIDELY FROM ONE CAB TO THE NEXT. WHY SUCH HIGH VARIABILITY? BECAUSE CABS—AND CABBIES—HAVE DIFFERENT PERSONALITIES!



A FAR BETTER WAY TO DO THIS STUDY IS TO ASSIGN GAS A AND GAS B TO THE SAME CAB ON DIFFERENT DAYS.



WE STILL RANDOMIZE THE TREATMENT BY FLIPPING A COIN TO DECIDE WHETHER TO USE GAS A ON TUESDAY OR WEDNESDAY. WE CAN ALSO CUT THE EXPERIMENT DOWN TO 10 CABS, SAVING THE OWNER A LOT OF TIME AND MONEY!

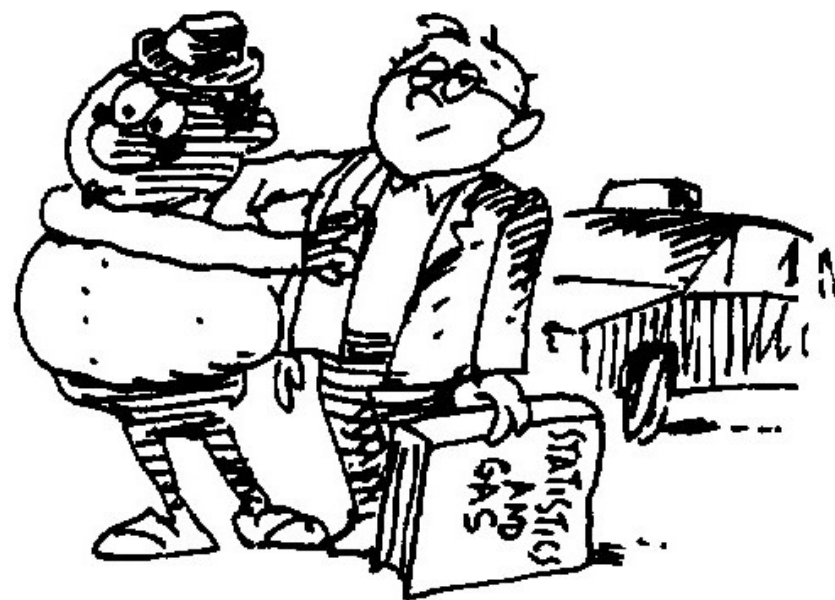


CAB	GAS A	GAS B	DIFFERENCE
1	27.01	26.95	0.06
2	20.00	20.44	-0.44
3	23.41	25.05	-1.64
4	25.22	26.32	-1.10
5	30.11	29.56	0.55
6	25.55	26.60	-1.05
7	22.23	22.93	-0.70
8	19.78	20.23	-0.45
9	33.45	33.95	-0.50
10	25.22	26.01	-0.79
MEAN	25.20	25.80	-0.60
STANDARD DEVIATION	4.27	4.10	0.61

NOTE THAT THE MEANS AND STANDARD DEVIATIONS OF GAS A AND GAS B ARE ABOUT THE SAME. THAT'S TO BE EXPECTED, SINCE THEY HAVE THE SAME SOURCE OF VARIABILITY AS IN THE UNPAIRED EXPERIMENT. BUT NOW THE DIFFERENCE COLUMN HAS A VERY SMALL STANDARD DEVIATION. THE DIFFERENCE COLUMN, BY COMPARING GAS PERFORMANCE WITHIN A SINGLE CAR, ELIMINATES VARIABILITY BETWEEN TAXIS.

THE DIFFERENCES d_i PROVIDE A SINGLE MEASURE OF DIFFERENCE FOR EACH TAXI, AND WE CAN USE IT TO MAKE A SMALL-SAMPLE t TEST STATISTIC:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}$$

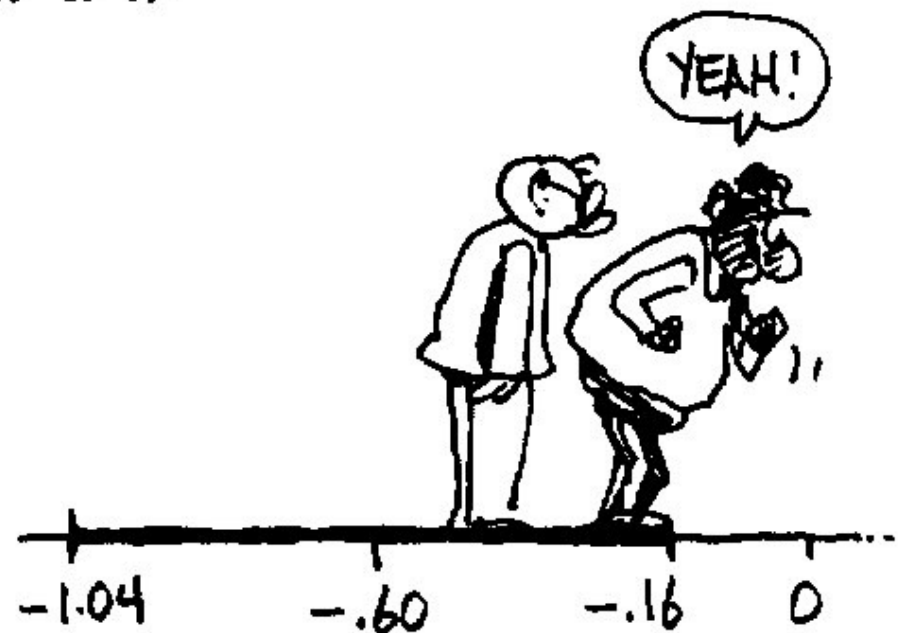


THE 95% CONFIDENCE INTERVAL AROUND $\bar{d}$ IS

$$\mu_d = \bar{d} \pm t_{.025} (s_d / \sqrt{n})$$

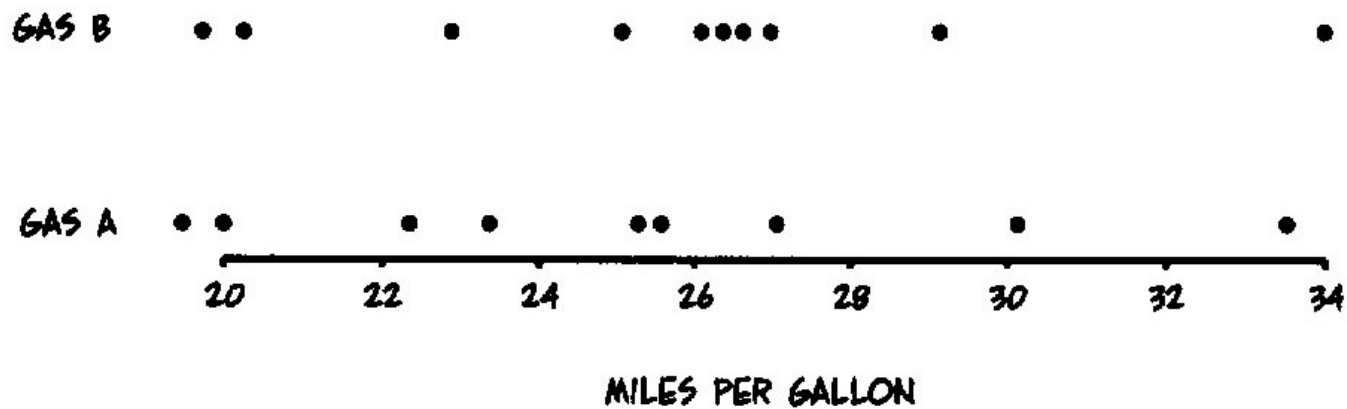
↑ ↑ ↑
SAMPLE MEAN CRITICAL VALUE STANDARD ERROR

$$= -.6 \pm (2.26) \left(\frac{.61}{\sqrt{10}} \right)$$
$$= -.60 \pm .44$$

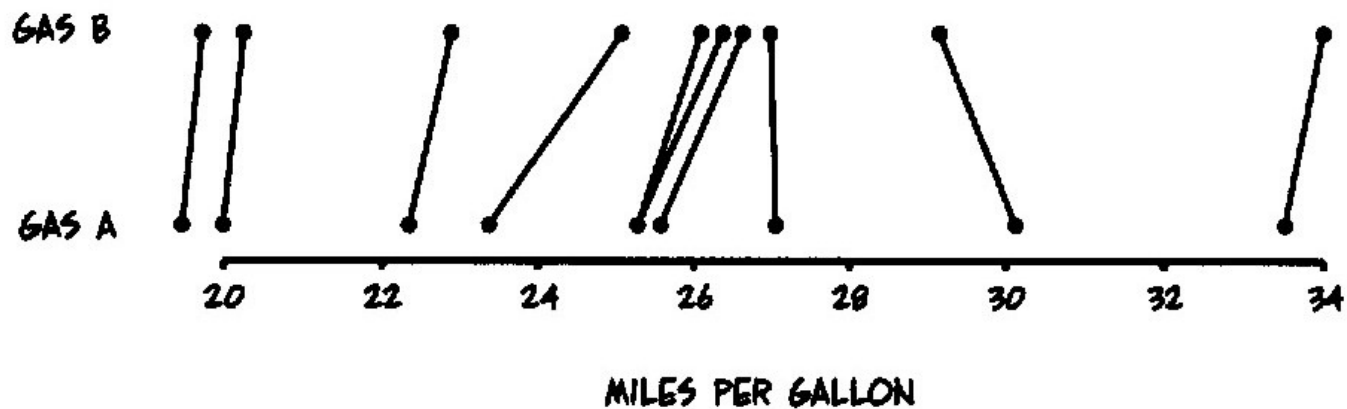


SO WE HAVE $-1.04 \leq \mu_d \leq -.16$ WITH 95% CONFIDENCE, GOOD EVIDENCE THAT GAS B REALLY IS BETTER.

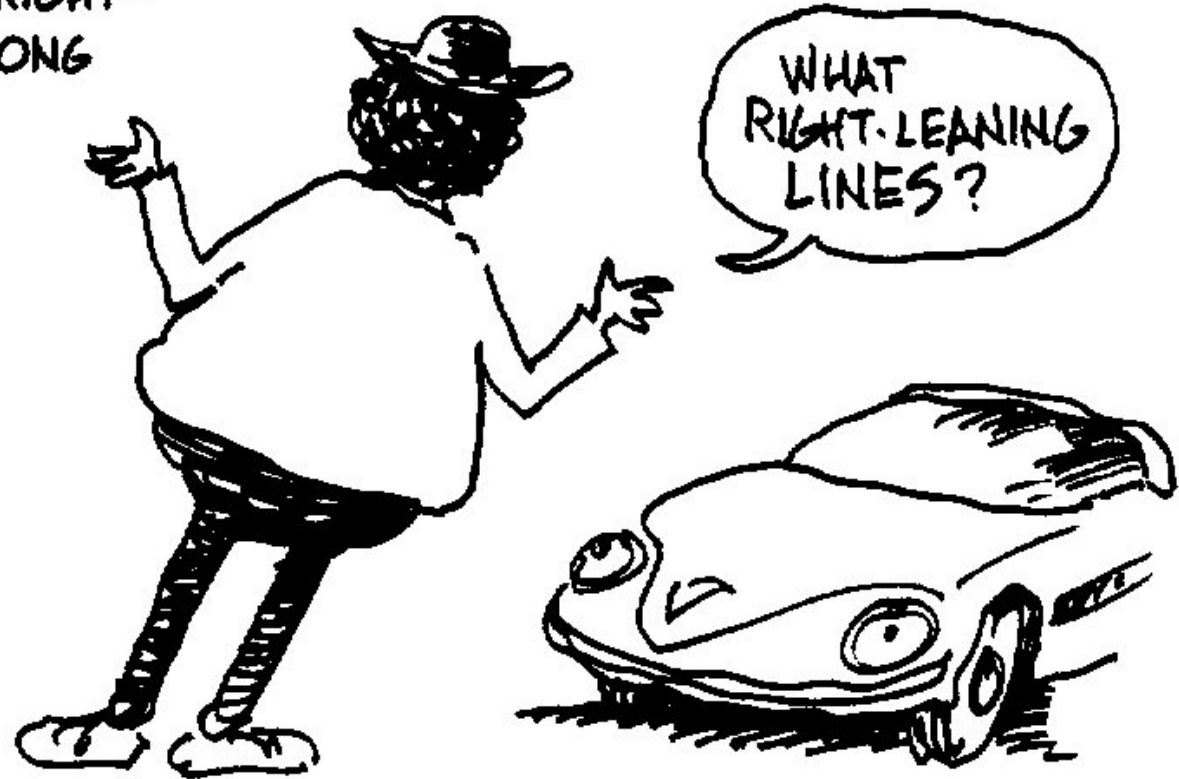
HERE ARE PLOTS OF THE GAS MILEAGE DATA: THE FIRST ONE SHOWS THE MILEAGES UNPAIRED:



AND HERE'S THE SAME DATA PAIRED BY TAXICAB.



THE PREDOMINANCE OF RIGHT-LEANING LINES IS A STRONG HINT THAT GAS B GIVES BETTER MILEAGE.



The key difference is:

The standard error between the two data sets!



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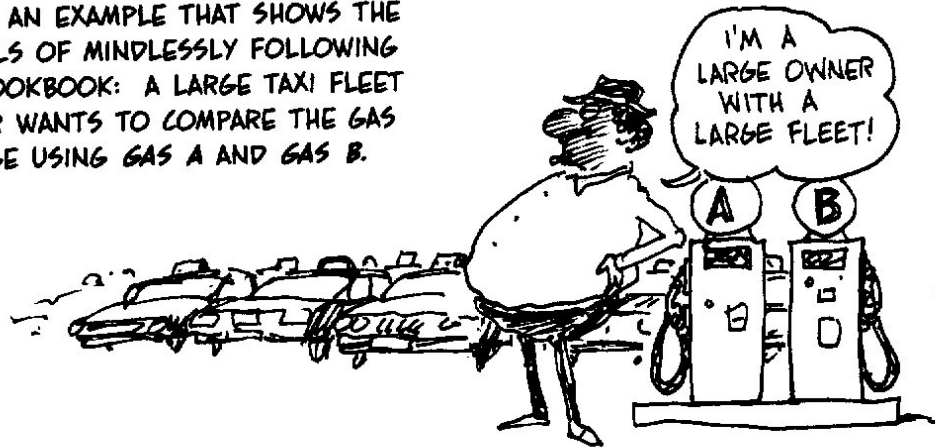
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 &= \sqrt{\frac{25}{50} + \frac{16}{50}} \\
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 \end{aligned}$$

STANDARD ERROR $(s_d / \sqrt{n}) = \left(\frac{.61}{\sqrt{10}} \right) = 0.193$

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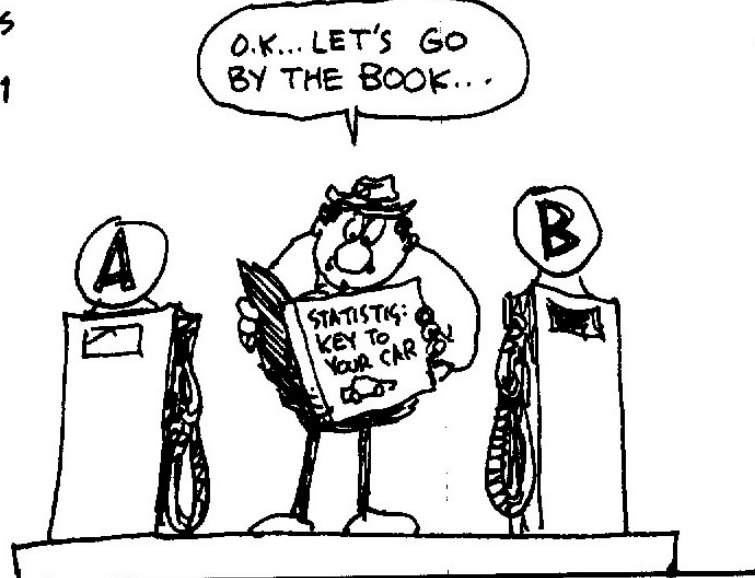
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STANDARD ERROR $(s_d / \sqrt{n}) = \left(\frac{.61}{\sqrt{10}} \right) = 0.193$

Why such a small standard deviation?

Paired data

X_i and Y_i are positively related for all $i=1,\dots,n$.

$V(X-Y)=?$

Two-sample data

X_i and Y_i are assumed to be independent for all $i=1,\dots,n$.

$\text{corr}(X_i, Y_i) = \rho = 0$.

$$V(X - Y) = \sigma_1^2 + \sigma_2^2$$

$$V(\bar{X} - \bar{Y}) = \frac{\sigma_1^2 + \sigma_2^2}{n}$$

$S_{\bar{X}-\bar{Y}}^2$: an estimator of $V(\bar{X} - \bar{Y})$

Variance of a linear combination of two random variables X and Y

$$V(aX+bY) = a^2V(X) + b^2V(Y) + 2ab\text{Cov}(X,Y)$$

In particular:

$$V(X+Y) = V(X) + V(Y) + 2\text{Cov}(X,Y)$$

$$V(X-Y) = V(X) + V(Y) - 2\text{Cov}(X,Y)$$

Covariance

The covariance between two random variables X and Y is:

$$\text{Cov}(X,Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

$$= \begin{cases} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y) f(x, y) dy dx & \text{continuous}(X, Y) \\ \sum_x \sum_y (x - \mu_X)(y - \mu_Y) p(x, y) & \text{discrete } (X, Y) \end{cases}$$

Proposition: $\text{Cov}(X,Y) = E[XY] - \mu_X \mu_Y$.

Intuition

If there is strong positive relation between X, Y :

if X increases, Y increases (probabilistically).

if X decreases, Y decreases (probabilistically).

If there is strong negative correlation between X, Y

if X increases, Y decreases (probabilistically).

Correlation

Definition: The correlation coefficient of X & Y, denoted by $\text{Corr}(X,Y)$, ρ_{XY} or ρ just defined by:

$$\text{Corr}(X,Y) = \rho_{XY} = \rho = \frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y}$$

$$-1 \leq \text{Corr}(X,Y) \leq 1.$$

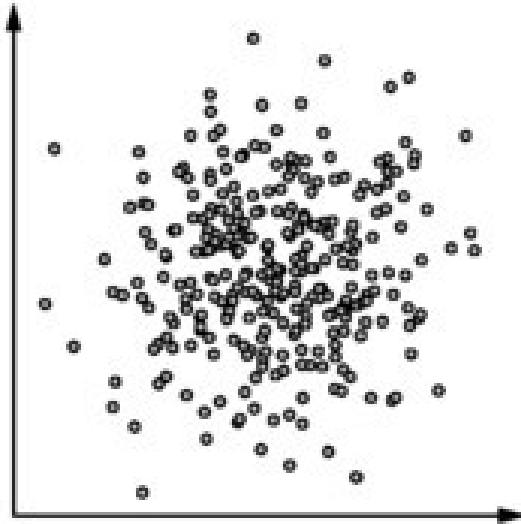
strong relation: if $|\rho| \geq 0.8$

moderate relation: if $0.5 \leq |\rho| < 0.8$

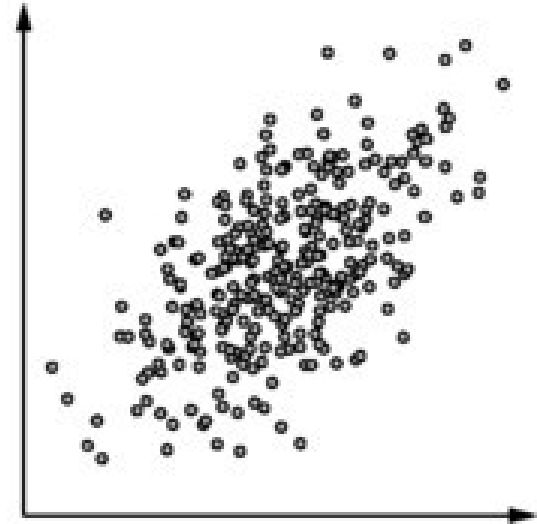
weak relation: if $|\rho| < 0.5$

Intuition

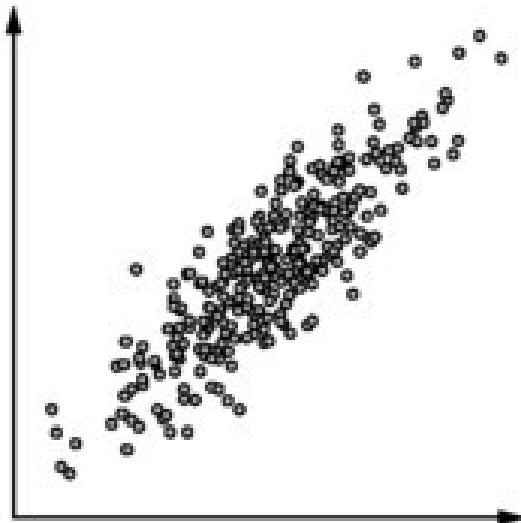
no
correlation



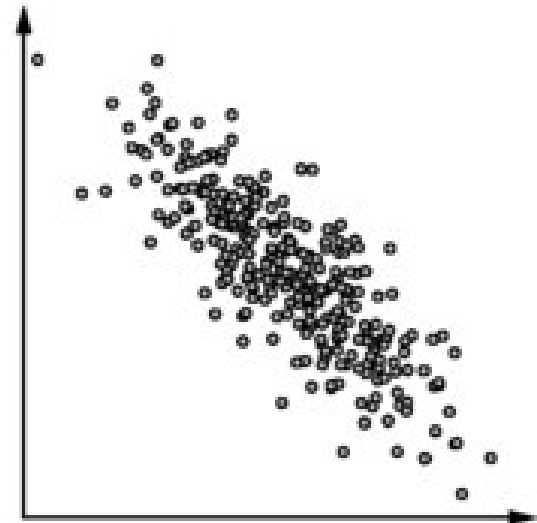
weak
positive
correlation



strong
positive
correlation



strong
negative
correlation



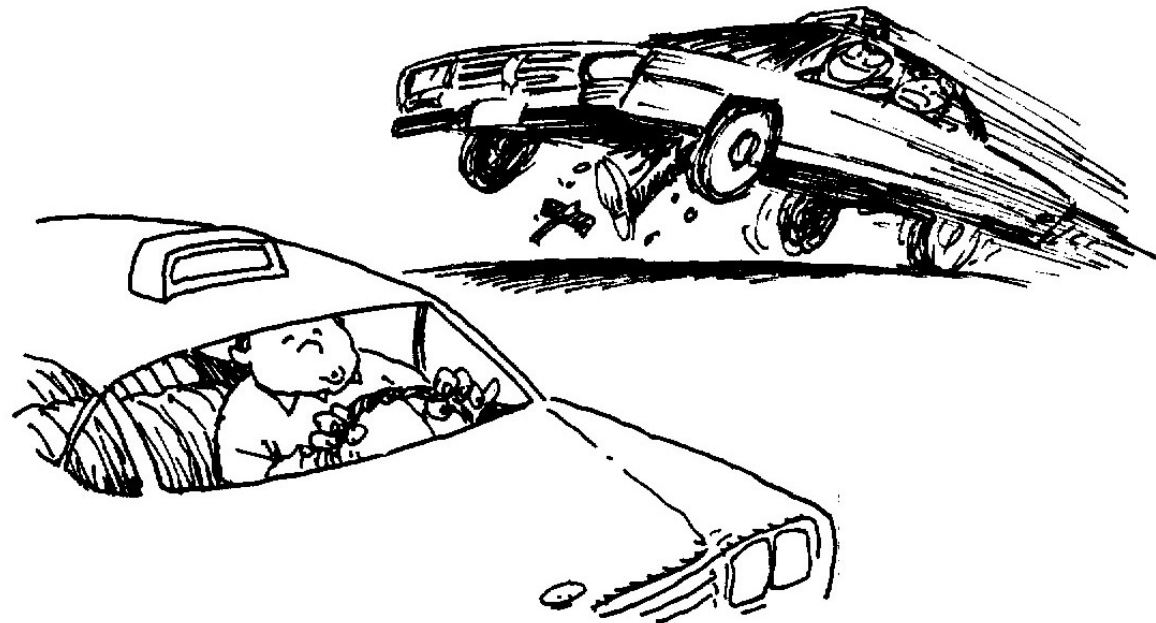
PAIRED COMPARISONS

a better way to compare gasolines



THE TAXI OWNER FOLLOWED THE COOKBOOK EXACTLY. HIS SAMPLES WERE RANDOM, AND HIS SAMPLE SIZE WAS LARGE ENOUGH. HE JUST FAILED TO *THINK* WHEN NECESSARY!

ALTHOUGH GAS B APPEARS TO BE SLIGHTLY BETTER THAN GAS A, THE CONFIDENCE INTERVAL WAS WIDE BECAUSE OF THE LARGE STANDARD DEVIATIONS—I.E., THE *MILEAGES VARIED WIDELY FROM ONE CAB TO THE NEXT*. WHY SUCH HIGH VARIABILITY? BECAUSE CABS—AND CABBIES—HAVE DIFFERENT *PERSONALITIES!*



Why such a small standard deviation?

Paired data

X_i and Y_i are positively correlated for all $i=1,\dots,n$.

$$\text{corr}(X_i, Y_i) = \rho > 0.$$

$$V(X-Y)=?$$

$$V(\bar{X} - \bar{Y}) = V(\bar{D}) = ?$$

$$S_{\bar{X}-\bar{Y}}^2 = S_{\bar{D}}^2 = ?$$

Two-sample data

X_i and Y_i are assumed to be independent for all $i=1,\dots,n$.

$$\text{corr}(X_i, Y_i) = \rho = 0.$$

$$V(X - Y) = \sigma_1^2 + \sigma_2^2$$

$$V(\bar{X} - \bar{Y}) = \frac{\sigma_1^2 + \sigma_2^2}{n}$$

an estimator of $V(\bar{X} - \bar{Y}) = ?$

Drawback of paired CIs??

Two-sample t CI

Paired t CI

X_i and Y_i are positively correlated for all $i=1,\dots,n$.

$$V(\bar{D}) = \frac{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}{n}$$

$S_{\bar{D}}^2$: an estimator of $V(\bar{D})$

$$d.f. = n - 1$$

X_i and Y_i are assumed to be independent for all $i=1,\dots,n$.

$$V(\bar{X} - \bar{Y}) = \frac{\sigma_1^2 + \sigma_2^2}{n}$$

$S_{\bar{X}-\bar{Y}}^2$: an estimator of $V(\bar{X} - \bar{Y})$

$$d.f. = \frac{\frac{(S_1^2 + S_2^2)^2}{n^2}}{\frac{1}{n-1} \left(\frac{S_1^4 + S_2^4}{n^2} \right)} = (n-1) \left(1 + \frac{2S_1^2 S_2^2}{S_1^4 + S_2^4} \right)$$

$$S_{\bar{D}}^2 < S_{\bar{X}-\bar{Y}}^2 \quad \text{BUT} \quad d.f.(\bar{D}) < d.f.(\bar{X} - \bar{Y})$$

Example

TABLE 5.1 Depths of tread, in mm, for tires made of new and old material

	Car									
	1	2	3	4	5	6	7	8	9	10
New material	4.35	5.00	4.21	5.03	5.71	4.61	4.70	6.03	3.80	4.70
Old material	4.19	4.62	4.04	4.72	5.52	4.26	4.27	6.24	3.46	4.50
Difference	0.16	0.38	0.17	0.31	0.19	0.35	0.43	-0.21	0.34	0.20

$$\bar{D} = 0.232$$

$$s_D = 0.183$$

When to use paired?

1. Great heterogeneity, high correlation:
loss in df gained with increased precision
→ paired-t CI is superior to two-sample t CI
2. Relative homogeneity, moderate correlation:
loss in df cannot be gained with increased precision
→ two-sample t CI is superior to paired-t CI

Inferences about the true variance of a random variable X

- $X_1, \dots, X_n$: A random sample with $\text{Var}(X) = \sigma^2$
- We are looking for a random variable $Y = h(X_1, \dots, X_n; \sigma)$ whose distribution can be specified.

CIs for the variance and standard deviation of a normal random variable:

$X_1, \dots, X_n$: random sample

Assumptions:

- $X_i \sim N(\mu, \sigma)$

$$\longrightarrow \frac{(n-1)S^2}{\sigma^2} = \frac{\sum (X_i - \bar{X})^2}{\sigma^2} \sim \chi_{n-1}^2$$

Normality is crucial, CLT does not apply here!!!

Definition:

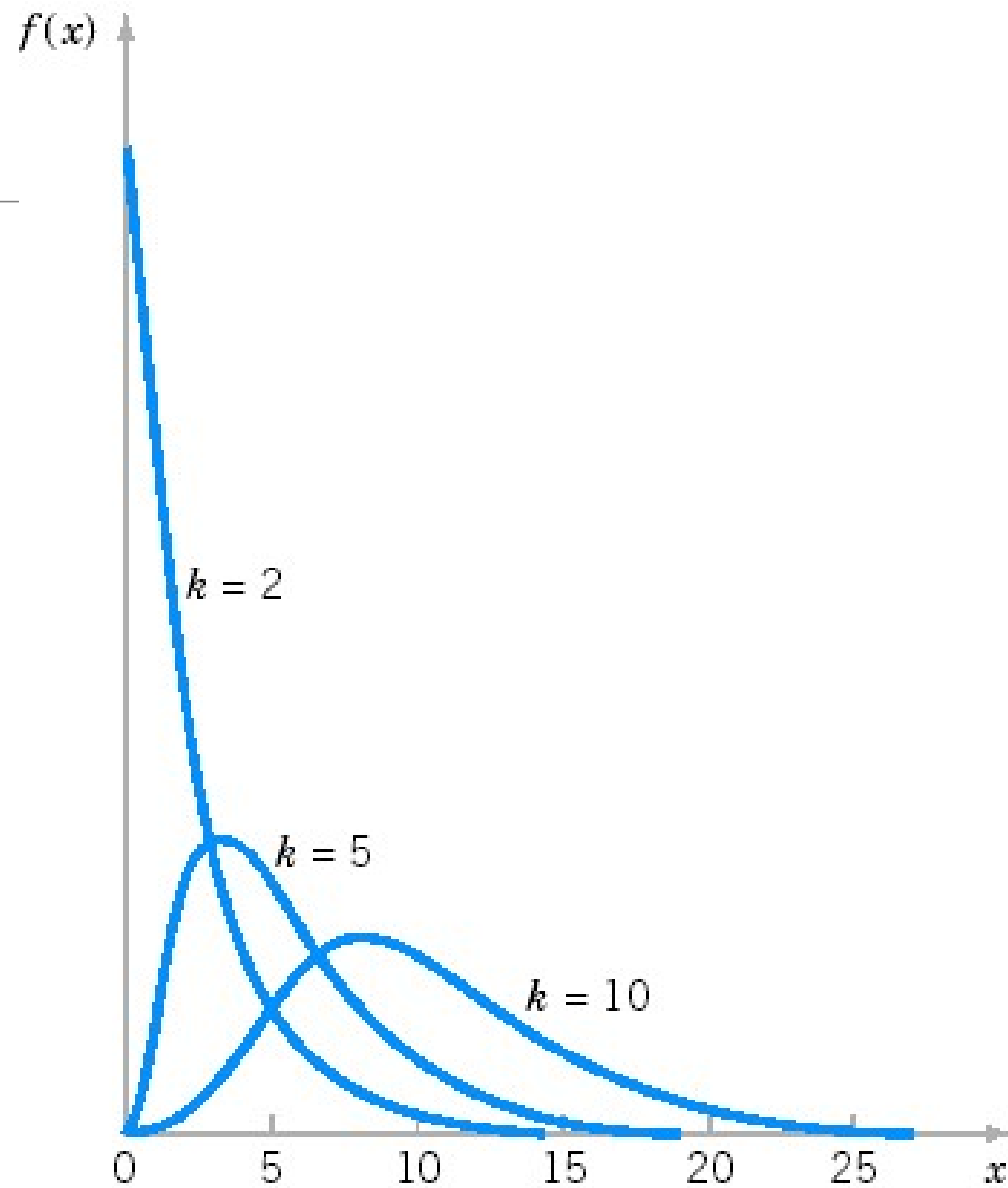
χ^2 Chi-Squared Distribution

Let ν be a positive integer. Then a random variable X is said to have a chi-squared distribution with parameter ν , and the pdf of X is given by:

$$f(x; \nu) = \begin{cases} \frac{1}{2^{\nu/2} \Gamma(\nu/2)} x^{(\nu/2)-1} e^{-x/2} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

The parameter ν is called the number of degrees of freedom (df) of X .

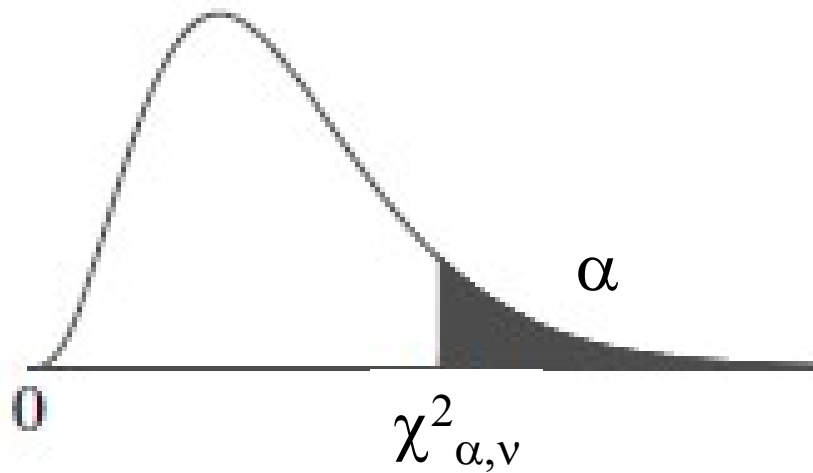
Probability density functions of several χ^2 distributions.



Notation

Difficult to compute, so the corresponding values are tabulated.

$\chi^2_{\alpha, v}$ is the critical value.



χ^2 -table

Degrees of Freedom	Chi-Square (χ^2) Distribution							
	Area to the Right of Critical Value							
	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01
1	—	0.001	0.004	0.016	2.706	3.841	5.024	6.635
2	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345
4	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277
5	0.554	0.831	1.145	1.610	9.236	11.071	12.833	15.086
6	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812
7	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475
8	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090
9	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666
10	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209
11	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725
12	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217
13	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688
14	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141
15	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578
16	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000
17	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409
18	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805
19	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191
20	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566
21	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932
22	9.542	10.982	12.338	14.042	30.813	33.924	36.781	40.289
23	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638
24	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980
25	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314
26	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642
27	12.879	14.573	16.151	18.114	36.741	40.113	43.194	46.963
28	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278
29	14.257	16.047	17.708	19.768	39.087	42.557	45.722	49.588
30	14.954	16.791	18.493	20.599	40.256	43.773	46.979	50.892

CIs for the variance and standard deviation of a normal random variable:

$X_1, \dots, X_n$: random sample

Assumptions:

- $X_i \sim N(\mu, \sigma)$

$$\longrightarrow \frac{(n-1)S^2}{\sigma^2} = \frac{\sum (X_i - \bar{X})^2}{\sigma^2} \sim \chi_{n-1}^2$$

Normality is crucial, CLT does not apply here!!!

Proposition

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$$

A confidence interval for the variance of normal random variable with confidence level $100(1-\alpha)\%$ has:

lower confidence limit: $\frac{(n-1)s^2}{\chi_{\alpha/2, n-1}^2}$

and

upper confidence limit: $\frac{(n-1)s^2}{\chi_{1-\alpha/2, n-1}^2}$

A confidence interval for σ has lower and upper limits that are square roots of the corresponding limits in the interval of σ^2 .

Example

How can we interpret these bounds?
Does the manufacturer achieve the desirable level for the standard deviation? Why or why not?

A manufacturer of soft drink beverages is interested in the uniformity of the machine used to fill cans. Specifically, it is desirable that the standard deviation σ of the filling process be less than 0.2 cm³; otherwise there will be a higher than allowable percentage of cans that are underfilled. We can assume that the fill volumes, X_i 's, are normally distributed. A random sample of 20 cans result in a sample variance of $s^2=0.0211$ cm³, so that $s=0.145$. Do the necessary analysis to find out about the confidence level of satisfying the need of $\sigma < 0.2$.

**Upper
confidence
bounds**

α	0.1	0.05	0.025	0.01
variance	0.0344	0.0396	0.045	0.0525
std. dev.	0.1855	0.1991	0.2122	0.2292

Why CI?

A random variable X distributed according to a pdf with an unknown parameter

Interest: To identify the unknown parameter

CI gives information on the unknown parameter of the distribution by using a random sample

Inferences about the true variance of a random variable X

$X_1, \dots, X_n$: A random sample

We are looking for a test statistics, $Y=h(X_1, \dots, X_n; \theta)$ such that its probability distribution does not depend on θ or any other unknown parameters.

$X \sim \text{Normal}$		X is not normal	
$n < 40$	$n \geq 40$	$n < 40$	$n \geq 40$
$\chi^2_{n-1} = \frac{(n-1)S^2}{\sigma^2}$	$\chi^2_{n-1} = \frac{(n-1)S^2}{\sigma^2}$	—	—